Roll No. BT23002OCS

BT/CS/W-25/CS5E001A/02/

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

Education to Eternity

B. Tech ~~Summer~~ / Winter - 2025

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: Computer Science & Engineering **Sem: - V**

Subject: PEC-I (Interactive Graphics) **Subject Code: CS5E001C**

Marks: 100 **Date of Exam:** 07/11/2025 **Time: 3 Hrs.**

Instructions to the students:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:

1. Demonstrate interactive graphics techniques using input devices.
2. EImplement 2D transformations, clipping & viewing pipelines.
3. Develop solutions using 2D geometric transformations.
4. Design interactive graphical systems with GUIs.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define the term Pixel and explain its role in raster graphics.	L1/CO1	2
Q.2	Identify the use of GUI.	L1/CO1	2
Q.3	List any two colour models used in computer graphics.	L2/CO2	2
Q.4	Define Plasma panel.	L1/CO2	2
Q.5	State the concept of digital frame buffer.	L2/CO2	2
Q.6	Define screen coordinates.	L2/CO3	2
Q.7	Distinguish between device coordinates and world coordinates.	L2/CO4	2
Q.8	Write two applications of circle drawing algorithm.	L3/CO4	2
Q.9	Identify the example of homogeneous coordinate.	L2/CO5	2
Q.10	Define reflection in 2D transformations.	L1/CO5	2
Q.11	Define polygon clipping	L1/CO5	2
Q.12	Define translation in 3D graphics and give its general matrix form.	L1/CO3	2

SECTION B

1. **All Questions are compulsory.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Distinguish between Raster and Vector display.	L4/CO1	4
Q.14	Illustrate colour display techniques with diagram.	L3/CO2	4
Q.15	Discuss compressed incremental list and vector list.	L2/CO2	4
Q.16	Examine the view algorithm in 2D transformations.	L4/CO3	4
Q.17	Write short note on clipping operations	L3/CO4	4
Q.18	Sketch the pipeline of 2D viewing transformation.	L3/CO5	4

SECTION C

1. **Solve any FOUR Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate line drawing algorithms with examples.	L4/CO2	7
Q.20	Explain circle drawing algorithm with step-by-step example.	L5/CO4	7
Q.21	Analyse hidden surface elimination methods in 3D graphics.	L4/CO5	7
Q.22	Examine text clipping and exterior clipping operations.	L4/CO5	7
Q.23	Describe projective transformations in 3D graphics.	L5/CO1	7
Q.24	Discuss GUI and interactive techniques in 3D graphics.	L5/CO3	7

SECTION D

1. **Solve any TWO Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Explain about following: a) Raster display b) Shadowmask CRT c) Colour look-up tables	L5/CO1, CO3, CO4	12
Q.26	Justify the importance of transformation pipeline in 2D graphics.	L6/CO4	12
Q.27	Design a 3D transformation model with an example.	L6/CO5	12

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